

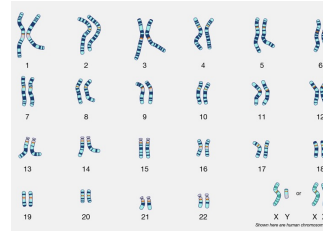
From 68 to 990 Genomes: A Conserved Chromosome-Size Relationship Persists

Meabh Hennessy^{1,4}, Peng Liu², Xianran Li³, Jianming Yu⁴

¹Carnegie Mellon University, Pittsburgh, PA; ²Department of Statistics, Iowa State University, Ames, IA; ³USDA-ARS Wheat Health, Genetics, and Quality Research Unit, Pullman, WA; ⁴Department of Agronomy, Iowa State University, Ames IA

Background and Research Question

- Chromosome number and length vary widely among eukaryotes, yet chromosome sizes within a species may follow a shared organizational pattern.
- Chromosome-size variation influences chromosome behavior during meiosis and organismal development (Schubert and Oud, 1997; Schubert, 2001).
- When modeling chromosomes from 68 eukaryotic genomes, Li et al. (2011) identified a conserved boundary of chromosome-size variation.
- Since 2011, many additional genomes have become available.
- Research Question:** How does the conserved chromosome-size relationship persist across a much larger dataset?



Human karyotype illustrating variation in chromosome size. Courtesy: National Human Genome Research Institute.

Materials & Methods

Filtering process for final chromosome assemblies:

2025 NCBI RefSeq assemblies $\Rightarrow$ Retain complete nuclear chromosome sets $\Rightarrow$ Exclude sex chromosomes $\Rightarrow$ **Final Dataset:** 990 eukaryotic genomes and 19,019 chromosomes.

Following Li et al. (2011), chromosome rank and length were standardized to allow comparisons among species with different genome sizes and chromosome counts:

$$\text{chromosome index} = \frac{\text{chromosome rank} - 0.5}{\text{chromosome number}}, \text{ relative chromosome size} = \frac{\text{chromosome length}}{\text{species mean chromosome length}}.$$

A gamma-quantile curve was fitted to predict relative chromosome size from chromosome index using iteratively reweighted least squares:

$$\text{relative chromosome size} = \frac{G^{-1}(\text{chromosome index}; \alpha, 1)}{\alpha}$$

where G^{-1} is the gamma quantile function, α is the fitted shape parameter, and $1/\alpha$ sets the mean to 1.

Results & Analysis

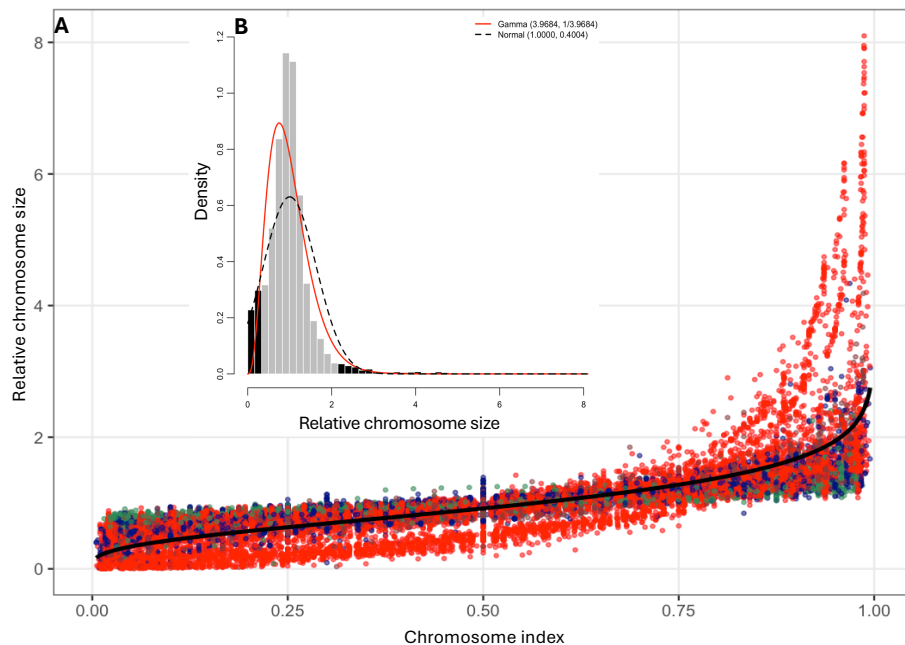


Figure 1. Chromosome-size patterns across 990 eukaryotic species.

(A) Relative chromosome size plotted against chromosome index with the fitted gamma-quantile curve ($\alpha = 3.968$).

(B) Observed distribution of relative chromosome size compared with the fitted gamma and normal reference distributions. The fitted model predicts a central 95% interval of **0.271–2.197**× the species mean.

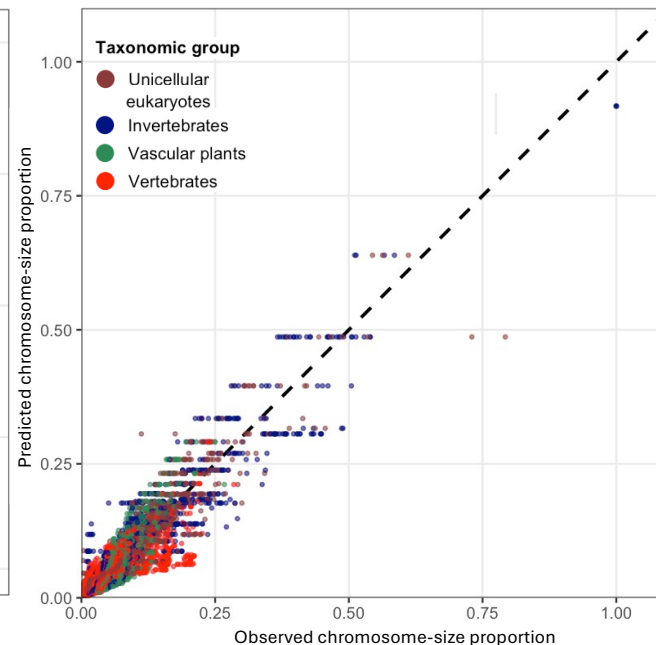


Figure 2. Agreement between observed and model-predicted chromosome-size proportions.

Chromosome-size proportion is chromosome length divided by the total retained chromosome length within each species. Predictions generally followed observed proportions, although deviations increased for chromosomes representing larger proportions of their species' genome.

Summary

Conclusions:

- Analysis of 990 genomes confirmed the conserved chromosome-size relationship reported by Li et al. (2011).
- The updated model predicted a wider central 95% boundary: 0.271–2.197 times the species mean, compared with 0.404–1.863 times in 2011.

Future Work:

- The same overall trend was visible across all major taxonomic groups. Differences between groups will be tested in future statistical analyses.

References

- Li X. et al. (2011). *Chromosome Size in Diploid Eukaryotic Species Centers on the Average Length with a Conserved Boundary*.
Schubert, I., & Oud, J. L. (1997). *There is an upper limit of chromosome size for normal development of an organism*.
Schubert, I. (2001). *Alteration of chromosome numbers by generation of minichromosomes—is there a lower limit of chromosome size for stable segregation*.
National Human Genome Research Institute. (n.d.). *Karyotype*. Genome.gov. Accessed July 28, 2026.

Acknowledgements

NCBI RefSeq data-processing workflow was developed by Dr. Leila Fattel.